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Date	30September 2026
TeamID	6ab4e2eba238dc999eb5fb00
ProjectName	**Script-Controlled ACL – Restrict Record Access Based on Field Value**
Maximum Marks	3 Marks

Script-Controlled ACL – Restrict Record Access Based on Field Value

Project Overview

This project demonstrates how **ServiceNow Access Control Lists (ACLs)** can be used to restrict record-level access based on a field value.

A custom **Institution Details** table is created in a ServiceNow Developer Instance. The table contains student and institutional information, including a **Branch** field with the choices **ECE, EEE and CSC**.

The main security requirement is to restrict the configured project user so that the user can access records belonging to the authorized **EEE branch** while records belonging to other branches are restricted.

Objectives

- Create a ServiceNow project user.
- Create the required roles: **BB1, BB2, BB3 and BB4**.
- Create the **Institution Details** table.
- Configure the required table fields.
- Configure ECE, EEE and CSC branch choices.
- Create sample records for testing.
- Implement Read, Create, Write and Delete ACLs.
- Use a script-controlled Read ACL based on the Branch field.
- Test authorized and restricted record access.
- Document the implementation with screenshots and test evidence.

Technologies and Platform

Item	Details

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Platform	ServiceNow Developer Instance
Security Mechanism	Access Control Lists (ACLs)
Main Table	Institution Details
Access Method	Role-based and field-based authorization
Script	ServiceNow ACL authorization script
Branch Choices	ECE, EEE, CSC

Institution Details Table

The project uses a custom table named **Institution Details**.

Main Fields

- Student ID
- Student Name
- Faculty Name
- Branch
- Email
- Phone Number
- Description

Branch Choices

- ECE
- EEE
- CSC

Sample Test Records

Student ID	Branch	Purpose
100	ECE	Test restricted access
200	EEE	Test authorized access
300	CSC	Test restricted access

Roles and ACL Matrix

Four roles are used to control the major operations.

ACL	Operation	Role	Purpose
ACL-01	Read	BB1	Read authorized Institution Details records
ACL-02	Create	BB2	Create Institution Details records
ACL-03	Write	BB3	Modify Institution Details records
ACL-04	Delete	BB4	Delete Institution Details records

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Script-Controlled Read ACL

The Read ACL is the main field-based security mechanism.

Configuration

- **Table:** Institution Details
- **Operation:** Read
- **Role:** BB1
- **Condition:** Branch is EEE

ACL Script

```
answer = (current.branch == 'EEE');
```

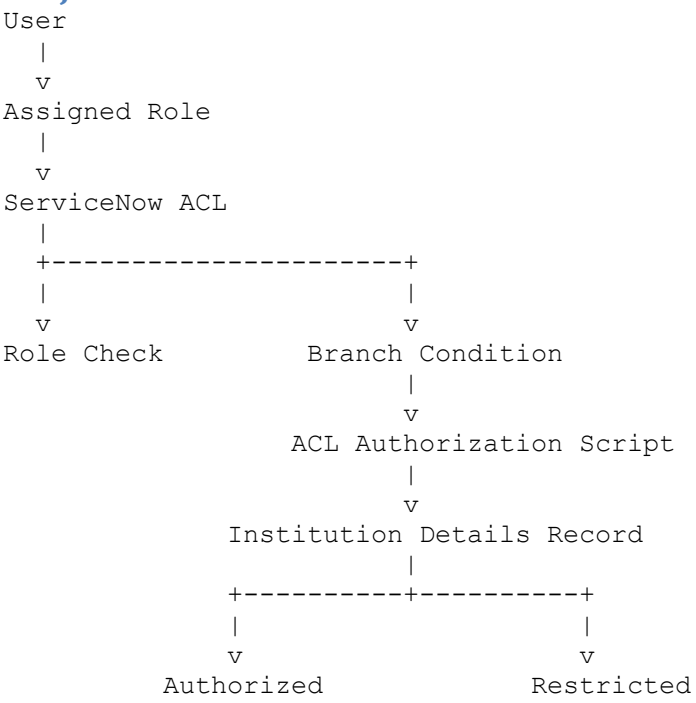
The script checks the Branch value of the current record.

If the Branch value is **EEE**, the authorization expression evaluates to `true`.

If the Branch value is not EEE, the record is restricted by the configured Read ACL.

Note: In ServiceNow, a Choice field's internal value may differ from its displayed label. The actual internal value configured for the EEE choice should be verified before using the script in a live instance.

Project Workflow



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System Operations

Read

The configured user with the required Read role is evaluated against the EEE branch condition.

Create

The Create ACL uses **BB2** to control creation of Institution Details records.

Write

The Write ACL uses **BB3** to control modification of records.

Delete

The Delete ACL uses **BB4** to control deletion of records.

Testing

The project includes testing for:

1. User creation
2. Role creation
3. Role assignment
4. Institution Details table configuration
5. Branch choice configuration
6. Sample record creation
7. EEE Read access
8. ECE Read restriction
9. CSC Read restriction
10. Create operation
11. Write operation
12. Delete operation
13. Unauthorized operation
14. ACL script validation
15. Final access validation
16. Administrator validation

Expected Read Result

Branch	Expected Result for EEE-restricted user
EEE	Access permitted
ECE	Access restricted
CSC	Access restricted

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Project Structure

A recommended GitHub repository structure is:

```
ServiceNow-Script-Controlled-ACL/
│
├── README.md
│
├── Documentation/
│   ├── Final_Project_Report.docx
│   ├── Problem_Statement.docx
│   ├── Requirements.docx
│   ├── ACL_Matrix.docx
│   ├── Team_Roles_and_Responsibilities.docx
│   └── Project_Schedule_7_Days.docx
│
├── Diagrams/
│   ├── Ideation_Diagram.png
│   ├── Architecture.png
│   ├── Use_Case.png
│   ├── DFD.png
│   └── ER_Diagram.png
│
├── Configuration/
│   ├── ACL_Script.txt
│   └── Update_Set.xml
│
├── Testing/
│   ├── Test_Cases.docx
│   └── Test_Screenshots.docx
│
└── Screenshots/
    ├── User_Creation.png
    ├── Table_Configuration.png
    ├── Sample_Records.png
    ├── Read_ACL.png
    ├── ACL_Script.png
    ├── Create_ACL.png
    ├── Write_ACL.png
    ├── ACL_Verification.png
    └── EEE_Access_Result.png
```

Team Roles

The project contains five members.

Member	Role	Responsibility
Member 1	Project Leader	Main project implementation, coordination, final submission and Google Drive video-link update
Member 2	Team Member	Documentation support
Member 3	Team Member	Project report preparation and formatting

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Member 4	Team Member	Testing, validation and screenshot evidence
Member 5	Team Member	Review, presentation/demo and submission support

The Project Leader is responsible for the final project submission and updating the project demonstration/video link in Google Drive.

7-Day Project Schedule

Day	Main Activity	Deliverable
Day 1	Project initiation and planning	Requirements and team responsibilities
Day 2	User, role and table creation	Users, roles and Institution Details table
Day 3	Branch configuration and sample data	ECE/EEE/CSC choices and sample records
Day 4	ACL implementation	Read, Create, Write and Delete ACLs
Day 5	Testing and validation	Test results and screenshots
Day 6	Documentation and video	Final report and demonstration video
Day 7	Final review and submission	Final project package and Google Drive link

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Project Deliverables

The project deliverables include:

- Problem Statement
- Requirements
- Ideation Diagram
- Architecture Diagram
- Use Case Diagram
- Data Flow Diagram
- ER Diagram
- Team Roles and Responsibilities
- 7-Day Project Schedule
- ACL Matrix
- ACL Script
- Update Set XML
- Test Cases
- Test Screenshots
- Final Project Report
- Project Demonstration Video
- Google Drive Video/Demo Link

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Expected Outcome

The completed project demonstrates field-based record-level security in ServiceNow.

The configured project user should be able to access the authorized **EEE** record while records belonging to **ECE** and **CSC** are restricted according to the Read ACL configuration.

The separate Create, Write and Delete ACLs provide operation-specific role-based control.

Conclusion

This project demonstrates a practical implementation of ServiceNow ACL security using roles, conditions and scripting. The combination of role-based permissions and a Branch-based Read restriction provides controlled access to Institution Details records.

The project also demonstrates the complete development and validation process, including user and role creation, table configuration, sample data creation, ACL implementation, script configuration, testing, documentation and final project submission.

Project Status

Project Type: ServiceNow Access Control Project
Implementation: Script-Controlled ACL
Main Security Rule: Restrict Read access based on Branch
Authorized Demonstration Branch: EEE
ACL Operations: Read, Create, Write, Delete
Testing: Completed / To be validated in the ServiceNow instance